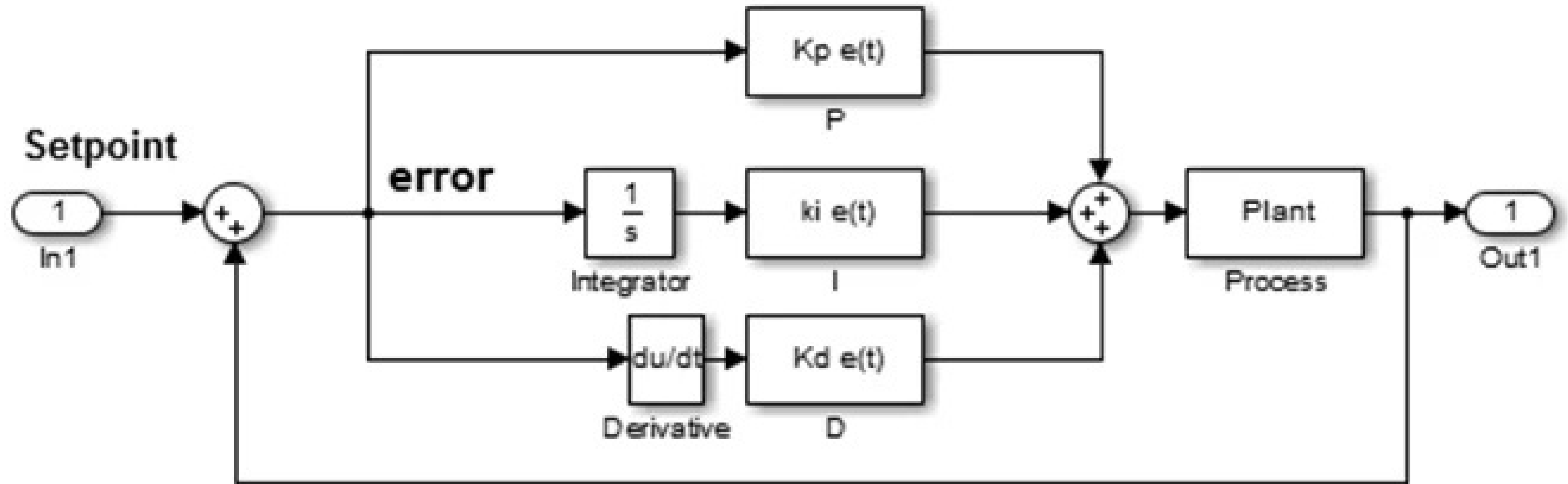


Dynamics and Controls - III

Mathematical Modelling of dynamical systems

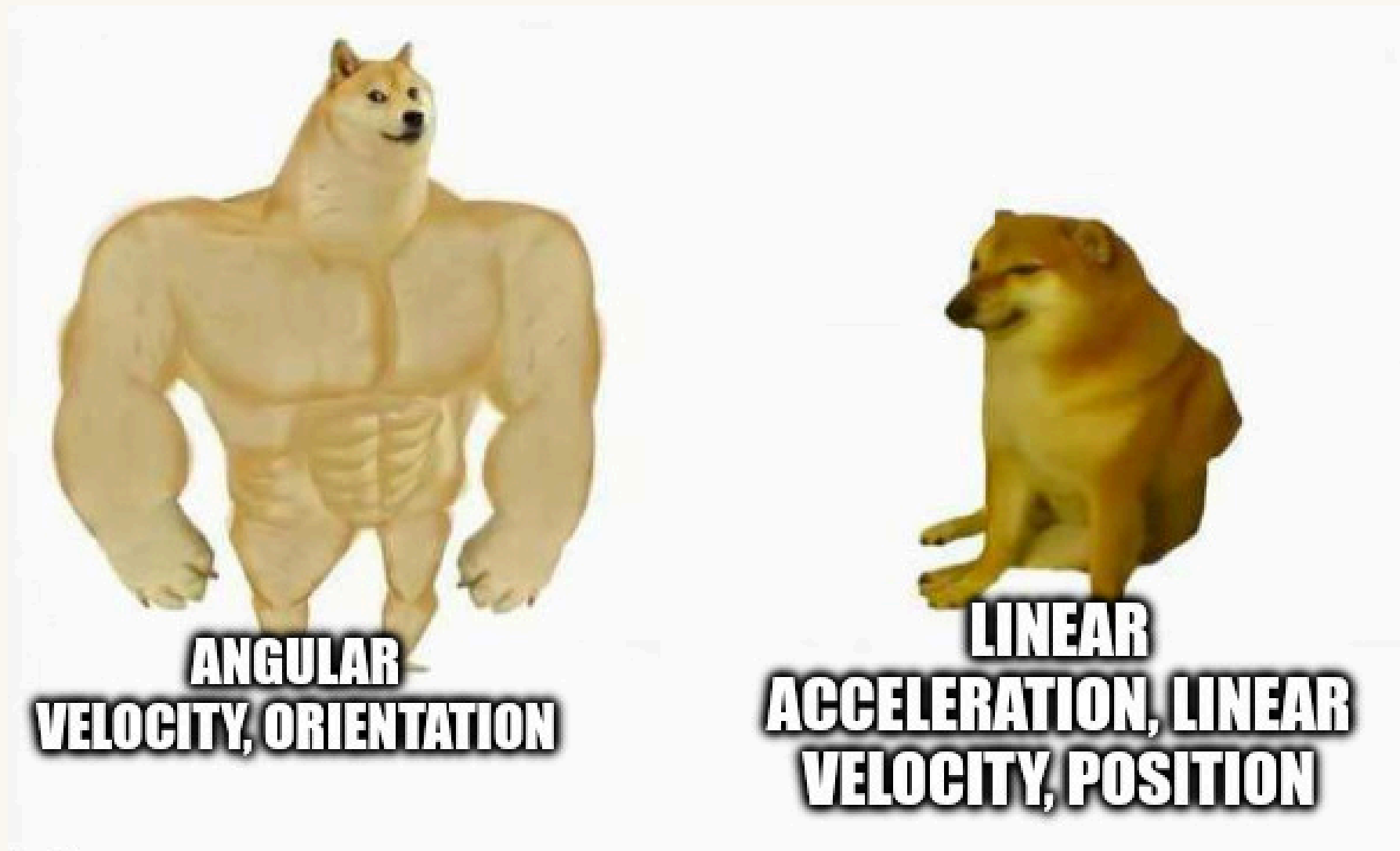
Astik Srivastava
Viswas Dinesh

Linear Systems: PID controller design

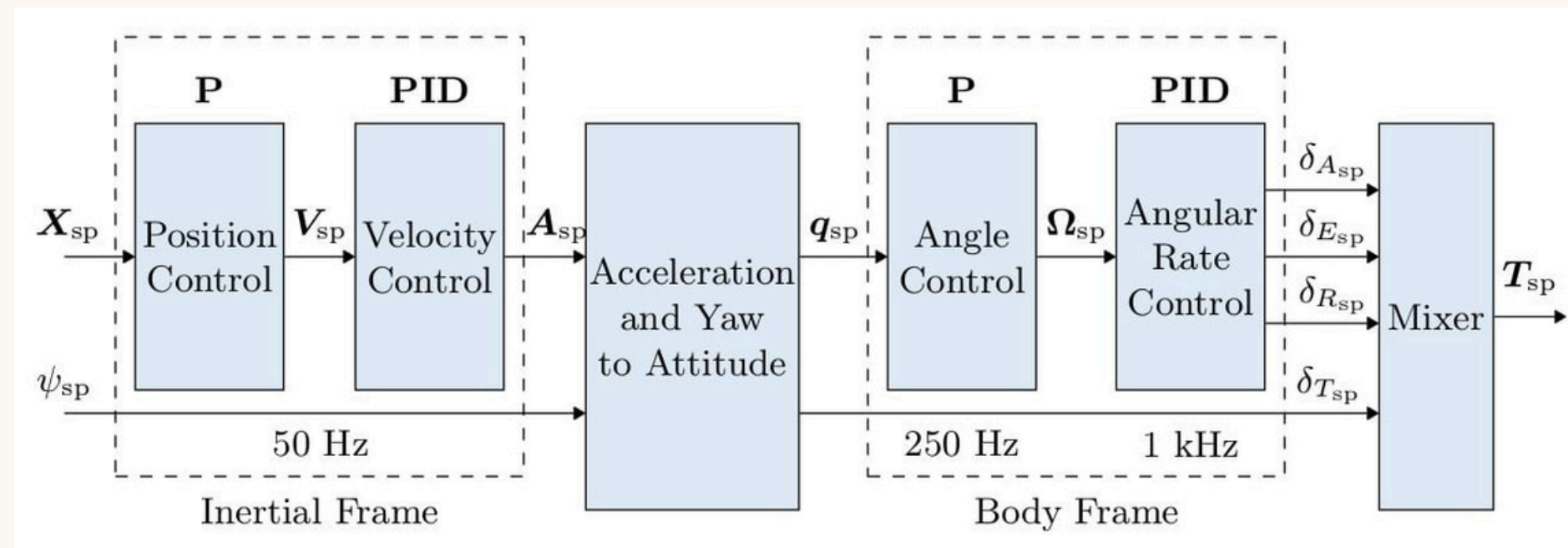


PID Controller

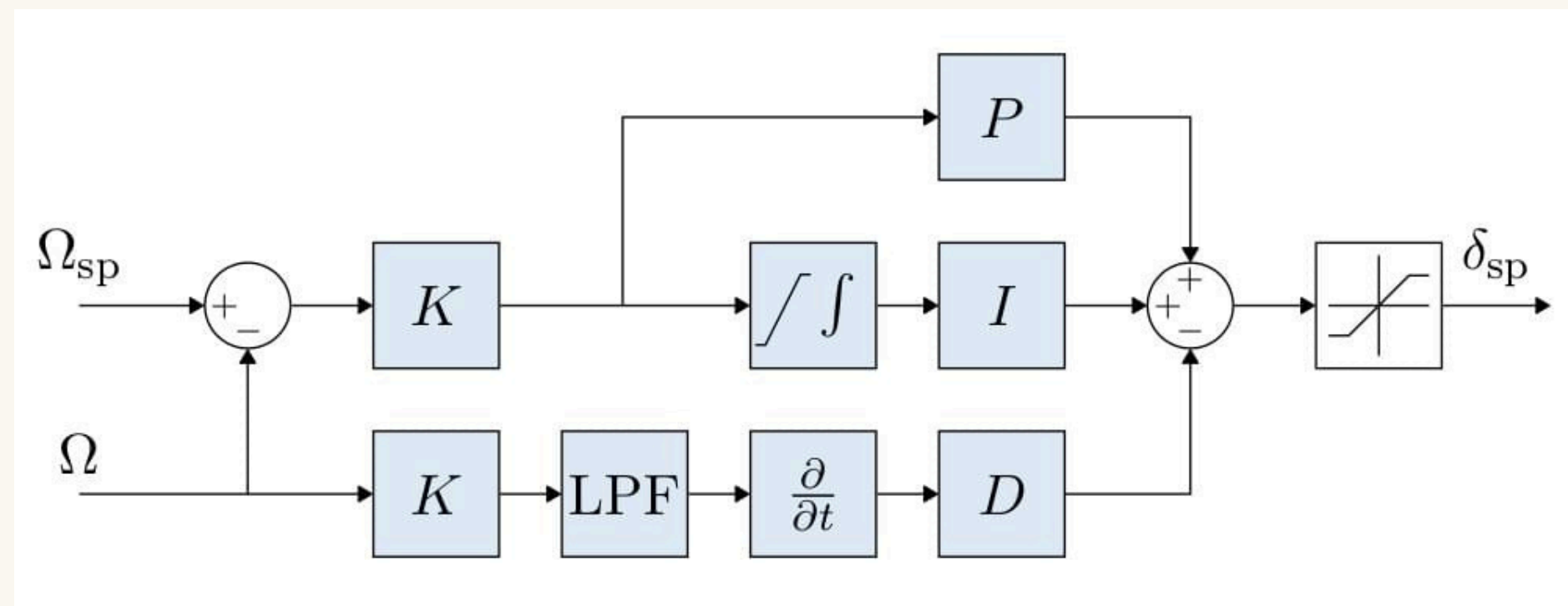
“Speed” of control of different states



PID Architecture Design



Cascaded PID

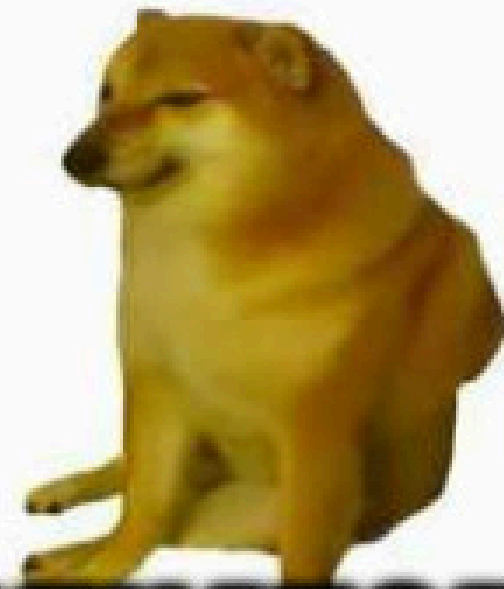


Individual PID loop

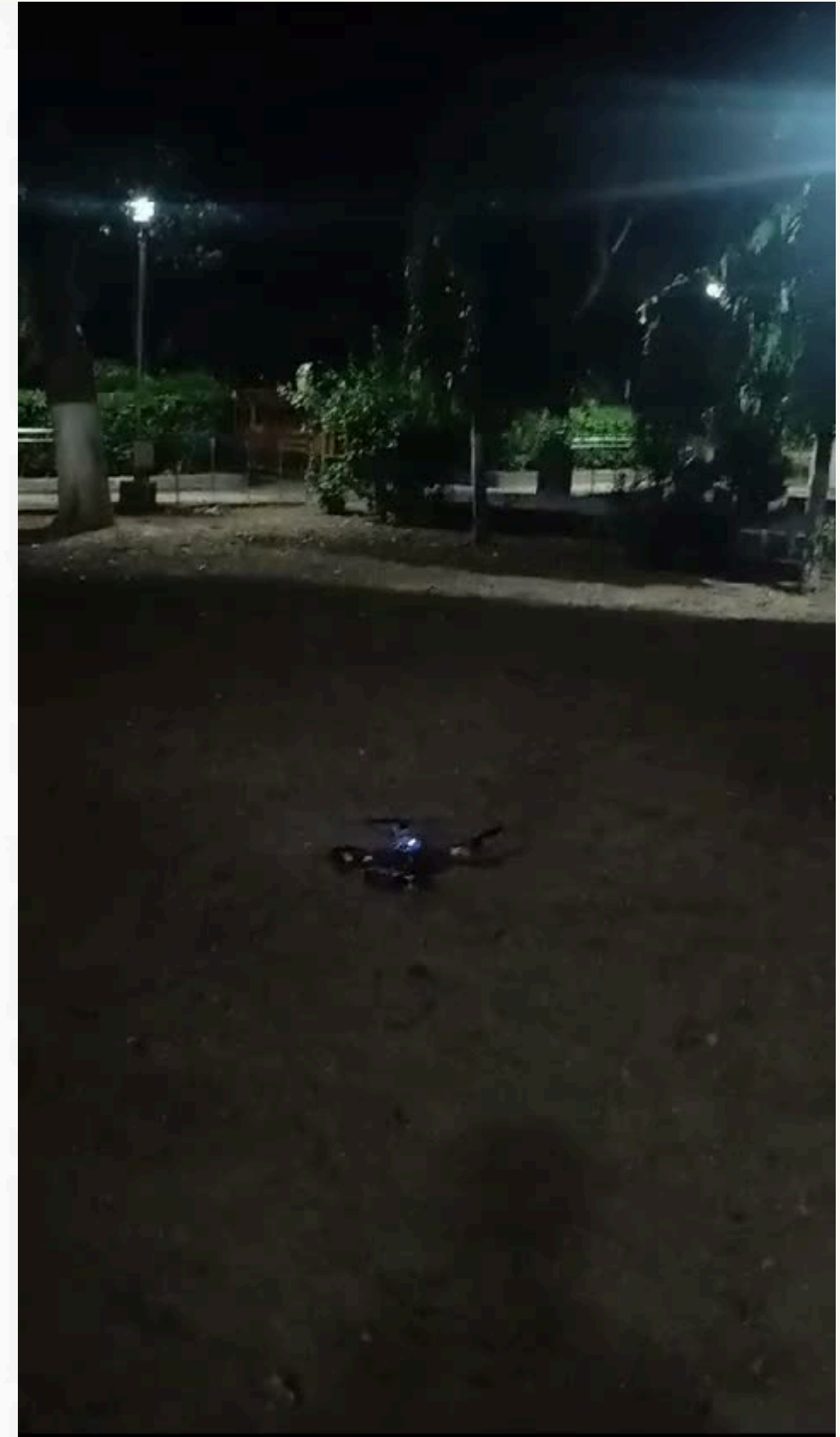
PID Architecture Design: Actuator Saturation



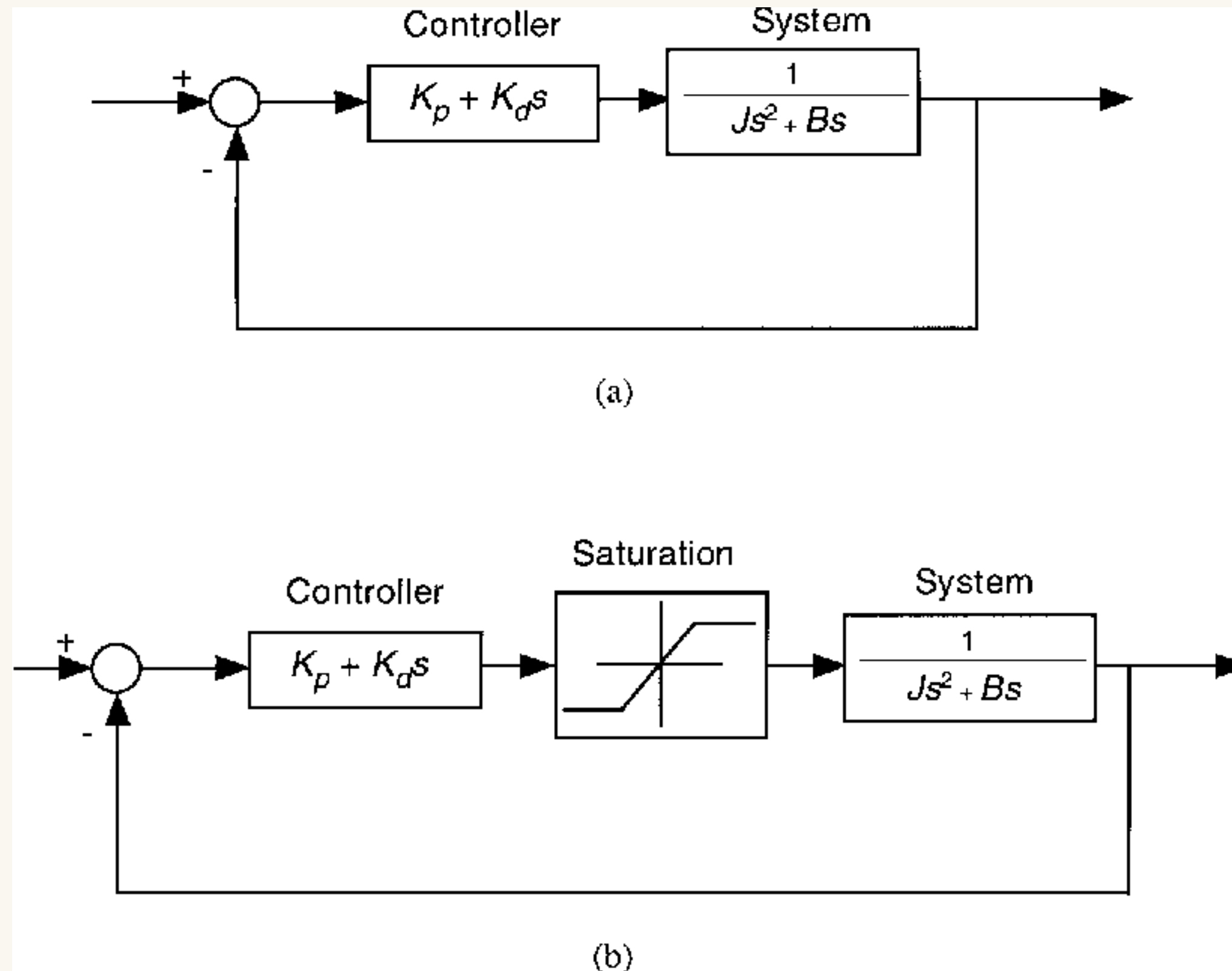
**ACTUATORS
IN SIMULATION**



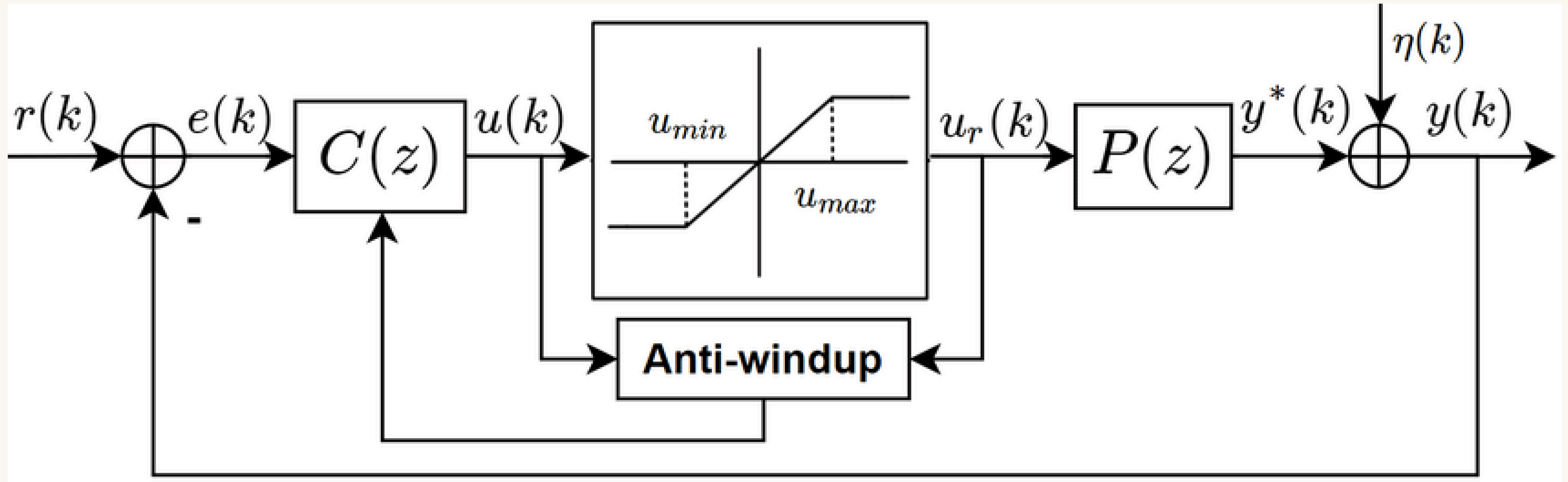
**ACTUATORS
IN REAL LIFE**



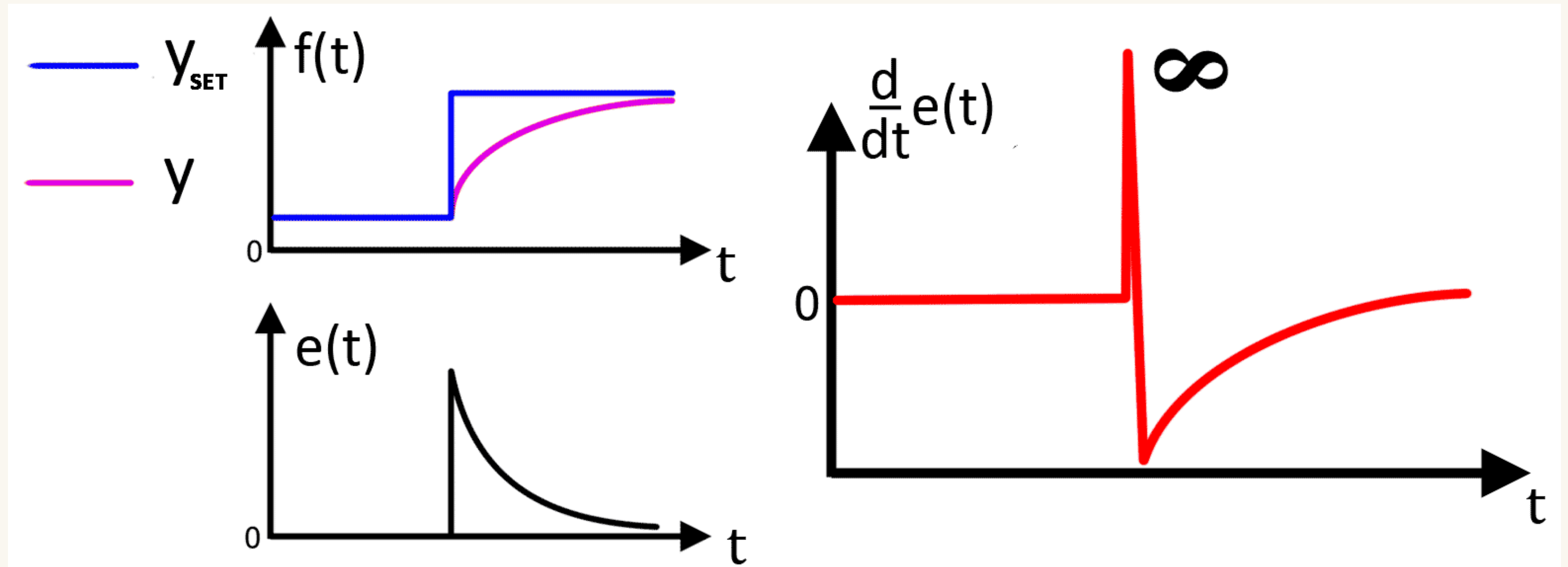
PID Architecture Design: Actuator Saturation



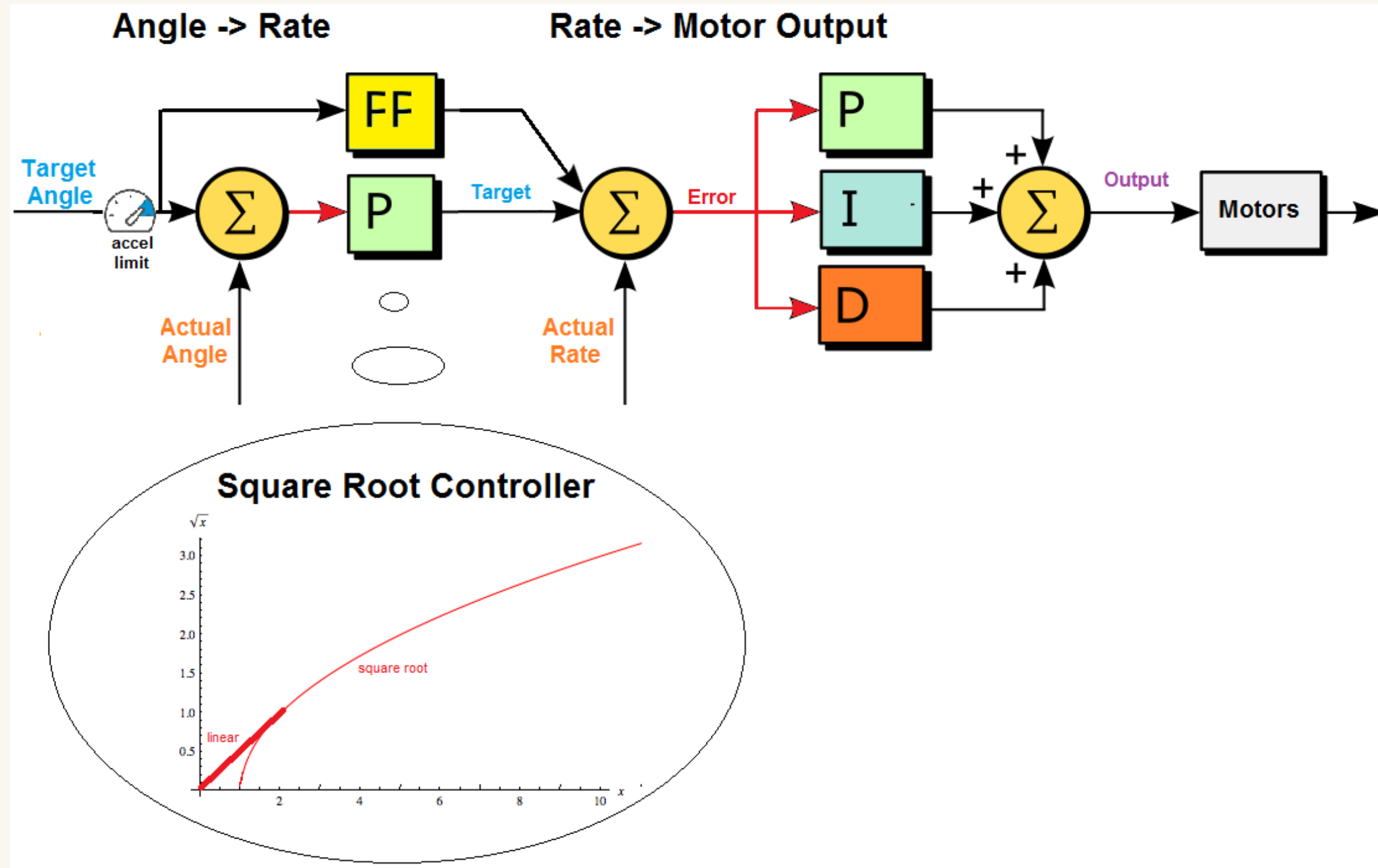
PID Architecture Design: Integral Windup



PID Architecture Design: Derivative kick



PID Architecture Design: Square Root Controller



Cautionary Tale for Controls



On the capstone project: Expectation



On the capstone project: Autonomous racing

